Sprint 0

Project Details	
Client	Global Mart Company
Project	Global Mart BI Project
Start Data	
End Date	
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Introductions

Problem Statement
A retail company sells bags, shoes, accessories, and other related products offline in their shop. The company has a record of all the orders in the year 2019, and 2020. The CEO of the company wants to know if the business is going in the right direction or not.

Stakeholders

Stakeholder Register							
Stakeholder Name	Role	Department	Contact Information	Influence	Interest	Engagement Strategy	Notes
John Doe	Project Sponsor	Executive Management	john.doe@ gm.com	High	High	Regular updates, major decisions	Key decision-maker, provides funding
Jane Smith	Product Owner	IT Department	jane.smith@ gm.com	High	High	Daily stand-ups, sprint reviews	Defines project vision and priorities
Mike Johnson	Data Analyst	Analytics Team	mike.johnso n@ gm.com	Medium	High	Weekly meetings, data workshops	Responsb-le for data modeling
Emma Brown	Marketing Manager	Marketing Department	emma.brow n@ gm.com	Low	Medium	Monthly updates, feedback sessions	Needs data for customer segmenta-tion
Alice Davis	Custome r	External	alice.davis@ gm.com	Medium	High	Surveys, feedback sessions	Provides user feedback
Regulatory Agency	Complian ce Officer	Government Body	contact@reg ulatory.gov	High	Medium	Quarterly compliance reports	Ensures project meets regulatory standards
Jira Stakeholder Register Link			https://alaa-hassan-project-				

	management.atlassian.net/jira/software/c/projects/GMSM/boards/4?atlOrigin=eyJpIjoiYjA5NmVmYjFlMDg1NDM2OTlmYWJhZTk1ZTg1YWM1NWIiLCJwIjoiaiJ9

Business Events

Business Event 1: Sales Transactions	
Stakeholder	Jane Smith (Product Owner)

Who does What	Customers purchases products	
7Ws	Who	Customers
	What	Purchase products
	When	Date & time of the transaction
	Event Type	Discrete (each sales transaction is a distinct, individual event)
	Where	Online stores in Brazil & Bangladesh.
	How many (Quantity & facts)	• number of products purchased
		• total sales amount
	Initial Data Example	• Customer A buys 2 units of product X of 3$ each from the Brazil store from Seller S on 2023-07-01 at 10:30 AM.
		• Customer B buys 1 unit of product Y of 2 $ from the Bangladesh store from Seller S on 2023-07-02 at 03:15 BM.
	Why	To buy products they need or want
	How	Through the e-commerce platform
	Fact Granularity	one single row for each product purchases per each transaction
	What else	• Payment method
		• promotion details
		• order status
Future Usage (External & Internal Data Sources)	• Predictive Sales Analysis • Market Basket Analysis	

Business Event 2: Inventory Updates	
Stakeholder	Mike Johnson (Data Analyst)

Who does What	Warehouse Staff/System updates inventory levels	
7Ws	Who	Warehouse Staff/System
	What	updates inventory levels
	When	Date & time of the update
	Event Type	Recurring (inventory updates happen periodically as sales and restocks occur)
	Where	Warehouses in Brazil and Bangladesh
	How many (Quantity & facts)	• number of units added to inventory. • number of units removed from inventory.
	Initial Data Example	• On **2023-07-01 at 11:00 AM**, the inventory for **Product X** in the **Brazil warehouse** is updated to reflect 50 units in stock after a restock of **30 units**. • On **2023-07-02 at 4:00 PM**, the inventory for **Product Y** in the **Bangladesh warehouse** is updated to reflect 20 units in stock after **10 units are sold**.
	Why	To reflect sales, returns, restocking.
	How	Through inventory management system.
	Fact Granularity	one single row for each inventory update at the level of individual product adjustments per warehouse.
	What else	• restock quantities • supplier details

Business Event 3: Customer Interactions	
Stakeholder	Alice Davis (Customer)

Who does What	Customer service Interaction.	
7Ws	Who	Customers
	What	Service Interaction.
	When	Date & time of the interaction.
	Event Type	Evolving (customer interactions).
	Where	Online stores in Brazil & Bangladesh
	How many (Quantity & facts)	• number of interactions
		• Interaction duration
	Initial Data Example	• On **2023-07-01 at 11:00 AM**, the Customer A service an inquiry about the shipment date of the order
		• On **2023-07-02 at 4:00 PM**, the inventory for **Product Y** in the **Bangladesh warehouse** is updated to reflect 20 units in stock after **10 units are sold**.
	Why	• Product Inquiry (Information Request) • Issue Resolution (Problem Solving) • Feedback (Customer Satisfaction)
	How	Through the different interaction channels (Physical store, Website, Phone, Email, Live Chat)
	Fact Granularity	One single row for each customer interaction.
	What else	
Future Usage (External & Internal Data Sources)	• Customer Sentiment Analysis	

Business Event 4: Marketing Campaigns	
Stakeholder	Emma Brown (Marketing Manager)

Who does What	Marketing Team/System launches and monitors marketing campaigns.	
7Ws	Who	Marketing Team/System
	What	Launches and monitors marketing campaigns.
	When	Date & time of the campaign launch and monitoring.
	Event Type	Evolving (campaigns are launched and evolve over time, with ongoing monitoring and adjustments).
	Where	Online stores in Brazil and Bangladesh, social media, email
	How many (Quantity & facts)	• Number of campaigns launched • Ad spend amount • Customer reach
	Initial Data Example	• On 2023-07-01 at 09:00 AM, a summer sale campaign is launched targeting customers in Brazil, with an ad spend of $500. • On 2023-07-02 at 10:00 AM, an email campaign promoting new arrivals is sent to customers in Bangladesh, with a target audience of 1,000 subscribers. • On 2023-07-03 at 11:00 AM, a social media ad campaign is launched to promote a flash sale in both Brazil and Bangladesh.
	Why	To increase sales, engage customers, promote products.
	How	Through marketing platforms and tools
	Fact Granularity	one single row for each marketing campaign event at the level of individual campaign actions and performance metrics.
	What else	• Campaign performance metrics. • Customer engagement.

Business Event 5: Financial Transactions	
Stakeholder	John Doe (Project Sponsor)

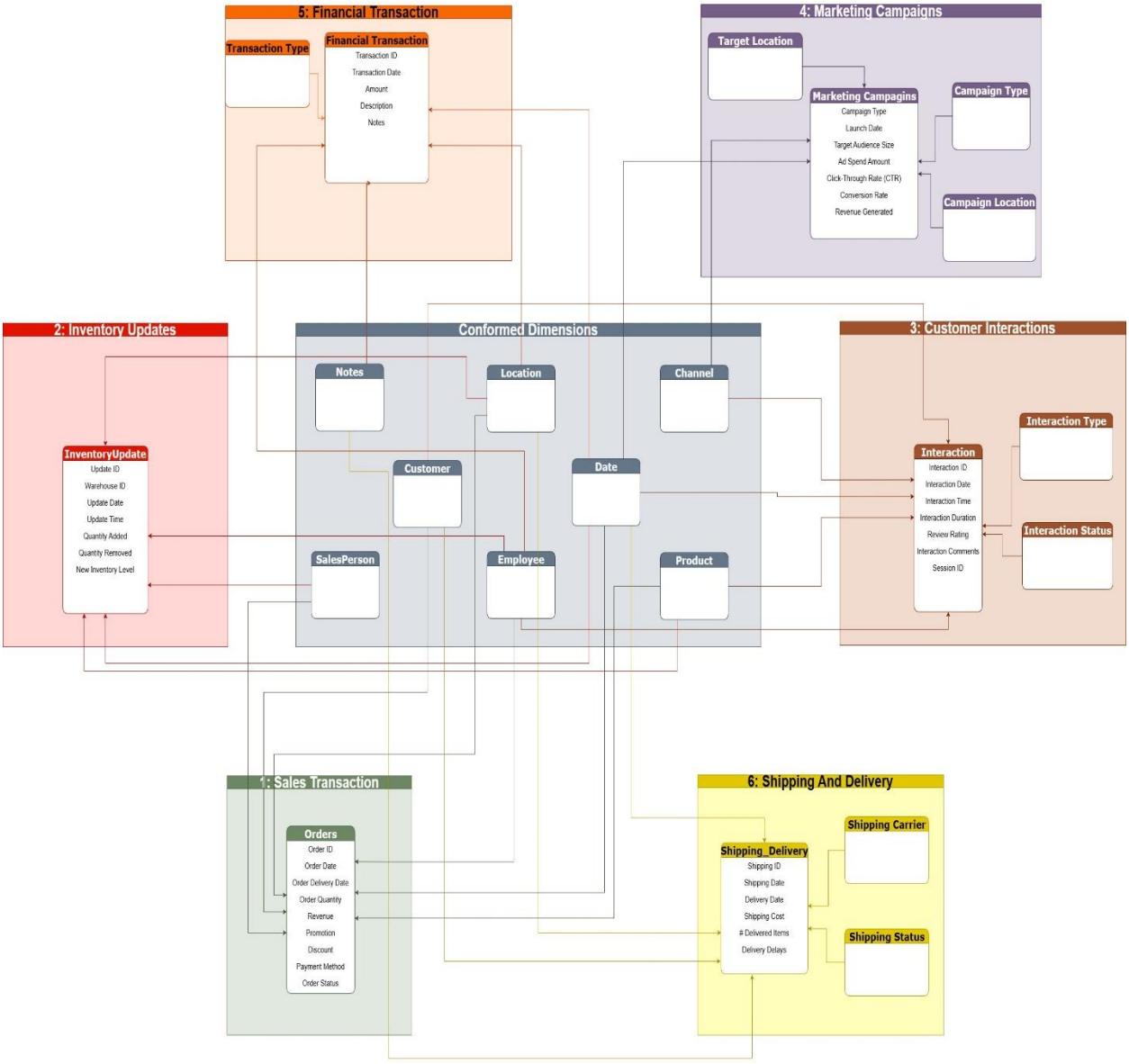
Who does What	Finance Team/System records financial transactions.	
7Ws	Who	Finance Team/System.
	What	Records financial transactions.
	When	Date & time of the transaction
	Event Type	Discrete (each financial transaction is a distinct, individual event).
	Where	Financial systems in Brazil & Bangladesh
	How many (Quantity & facts)	• Number of transactions
		• Total transaction amount
	Initial Data Example	• On 2023-07-01 at 02:00 PM, a revenue transaction of $1,000 is recorded for sales in the Brazil store.
		• On 2023-07-02 at 03:00 PM, an expense transaction of $200 is recorded for marketing spend in the Bangladesh store. • On 2023-07-03 at 04:00 PM, a budget allocation of $500 is recorded for the new marketing campaign in Brazil.
	Why	To track financial performance, manage budgets, and ensure compliance.
	How	Through financial management systems.
	Fact Granularity	one single row for each financial transaction at the level of individual entries in the financial system.
	What else	• Transaction details.
		• Account information

Business Event 6: Shipping & Delivery Event	
Stakeholder	Emma Brown (Logistics Manager)

Who does What	Logistics Managers/Shipping System records the shipping and delivery of orders.	
7Ws	Who	Logistics Team, Customer.
	What	Records the shipping and delivery details of orders.
	When	Date and time of the shipping and delivery events.
	Event Type	**Accumulating** (Each shipment and delivery event is monitored over time).
	Where	• Warehouses. • Stores. • shipping locations. • customer delivery addresses.
	How many (Quantity & facts)	• Number of shipments
		• Total shipment weight
		• Shipping cost
		• Number of delivered items
		• Delivery delays (if any)
	Initial Data Example	• On **2024-02-01 at 10:00 AM**, an order of **10 units** of **Product X** is shipped from the **Brazil warehouse** to **Customer A**, with a **shipping cost of $50**.
		• On **2024-03-15 at 02:30 PM**, an order of **5 units** of **Product Y** is delivered to **Customer B** in Bangladesh, weighing **15 kg**, with a delivery delay of **2 days**.
	Why	To track order shipments, optimize delivery performance, ensure on-time delivery, and manage shipping costs.
	How	Through logistics management systems integrated with data from shipping providers, warehouses, and customer addresses.
	Fact Granularity	one single row for each shipment event.
	What else	• Shipping Carrier Information
		• Delivery Delays & Performance Metrics (Tracking on-time deliveries vs.

		delayed deliveries for performance optimization)

Conceptual Model



Potential Data Sources

Data Sources & Data Ingestion Frequency

Data Sources					
Main Dataset		Data Sources		Data Ingestion Frequency	
		Source System	Source Type	Data Frequency	Timing Requirements
Brazilian E-Commerce		E-Commerce Platform	SQL Server	Daily	Near Real-Time (for updating sales data, inventory, etc)
Bangladesh E-Commerce	Bangladesh Ecommerce	E-Commerce Platform	MYSQL	Daily	Near Real-Time
	Customers	Customer Relationship Management (CRM) System	CSV	Periodic (weekly or monthly updates)	Batch processing (scheduled intervals)
	Items	Product Information Management (PIM) System	CSV		
	Stores	Store Management System	CSV		
	Transactions	Point of Sale (POS) System	CSV		
Import/Export Trade Data in Mexico		External	JSON	Periodic (depends on business expansion)	Batch Processing (scheduled intervals)
Retail Transactions Dataset		External	SQL Server	Ad-hoc (based on analysis needs)	Batch Processing (when running specific analysis)
Global Superstore		External	Script-Based (Python Environment)	Ad-hoc (based on analysis needs)	Batch Processing (when running specific analysis)

Data Quality & Transformation

Data Sources			
Main Dataset		Data Quality & Transformation	
		Data Quality Requirements	Data Transformation
Brazilian E-Commerce		High (consistent, accurate data is required for business operations)	High (Normalization, Aggregation, Cleaning, time-series transformation, feature engineering)
Bangladesh E-Commerce	Bangladesh Ecommerce	High (consistent, accurate data is required for business operations)	High (Normalization, Aggregation, Cleaning, time-series transformation, feature engineering)
	Customers	High (requires data validation and cleaning due to CSV format)	High (Agregation, Cleaning, feature engineering)
	Items		
	Stores		
	Transactions		
Import/Export Trade Data in Mexico		High (Parsing JSON, Text Processing)	Moderate (Parsing JSON, Aggregation, Text mining, feature extraction)
Retail Transactions Dataset		Medium (dependent on data source quality)	Moderate (Normalization, Aggregation, Cleaning)
Global Superstore		Medium (dependent on data source quality)	Moderate (Normalization, Aggregation, Cleaning)

Storage Requirements

Data Sources				
Main Dataset		Storage Requirement		
		Expected Data Volume		Storage Consideration
		Initially	Rate of change/year	
Brazilian E-Commerce		332.63 MB calculation	20%	Scalable storage in SQL Server (requires efficient indexing and partitioning for performance)
Bangladesh E-Commerce	Bangladesh Ecommerce	78.2MB	0.783%	Scalable storage in MYSQL Server (requires efficient indexing and partitioning for performance)
	Customers	772 KB	0.01%	Managed storage (#)
	Items	42.3 KB	0.01%	
	Stores	111.3 KB.	0.01%	
	Transactions	2.5 KB.	0.01%	
Import/Export Trade Data in Mexico		73KB	900%	Scalable NoSQL Database or JSON-compatible storage (e.g. MongoDB)
Retail Transactions Dataset		13.83 MB	20%	Scalable SQL Server storage with indexing and partitioning
Global Superstore		2.28 MB =>parquet 48.81MB=>pandas		Scalable storage in a Python environment with on-demand processing

Processing Requirements

Data Sources			
Main Dataset		Processing Requirements	
		Processing Type	Performance Requirements
Brazilian E-Commerce		Batch Processing with potential real-time updates for certain tables)	High (fast queries and transactions are essential for business operations)
Bangladesh E-Commerce	Bangladesh Ecommerce	Batch Processing with potential real-time updates	High (fast queries and transactions are essential for business operations)
	Customers	Batch Processing (periodic loads into the database)	Low to Medium (depends on frequency of use and load sizes)
	Items		
	Stores		
	Transactions		
Import/Export Trade Data in Mexico		Batch Processing, Text Mining	low-medium (depends on the complexity of analysis)
Retail Transactions Dataset		Batch Processing	Medium (focus on query optimization)
Global Superstore		Batch Processing	Medium-High (depending on the specific analysis and processing requirements)

Reporting Requirements

Data Sources				
Main Dataset		Reporting Requirements		
		Analysis Type	User Access and Security	
			User Roles & Access levels	Security Requirements
Brazilian E-Commerce		• Data Analysis • Reporting • Historical Trend Analysis • Predictive Analysis	- **Data Analysts**: Read Access (full data access for analysis) - **Business Managers**: Read/Write Access (access to aggregated reports and ability to trigger reports) - **IT Administrators**: Full Access (manage data ingestion and access control)	High (SQL Server security features, encryption, and access control)
Bangladesh E-Commerce	Bangladesh Ecommerce	• Data Analysis • Reporting • Historical Trend Analysis	- **Data Analysts**: Read Access (full access for trend and predictive analysis) - **Marketing Managers**: Read Access (access to key KPIs and sales analysis) - **Operations Managers**: Read/Write Access (access to inventory data and ability to generate reports)	High (MySQL security features, regular backups, access controls)
	Customers	• Data Analysis • Operational Reporting	- **Marketing Managers**: Read Access (access to customer segmentation data) - **Customer Service**: Read Access (basic data for customer support and issue tracking)	Medium (ensure file storage security, proper access control)
	Items		- **Inventory Managers**: Read Access (access to stock levels and product information) - **Data Analysts**: Read	

			Access (for analyzing product trends)	
	Stores		- **Store Managers**: Read Access (store-level sales and operational data) - **Regional Managers**: Read/Write Access (ability to aggregate data across stores)	
	Transactions		- **Finance Team**: Read Access (access to sales and financial data for reporting) - **Data Analysts**: Read Access (access to all transactional data for in-depth analysis)	
Import/Export Trade Data in Mexico		• Financial/Trade Analysis	- **Logistics Managers**: Read Access (access to shipping and trade data for optimizing supply chains) - **Business Managers**: Read Access (review trade analysis for decision-making)	Medium-High
Global Superstore		• Data Analysis • Historical Trend Analysis • Predictive Analysis	- **Data Scientists**: Full Access (for deep analysis and modeling) - **Business Managers**: Read Access (high-level KPI reports) - **Finance Team**: Read Access (sales and revenue analysis)	High (SQL Server security features, encryption, and access control) -high because competitive data-
Retail Transactions Dataset		• Financial and Sales Analysis	- **Data Scientists**: Full Access (on-demand access for analysis) - **Business Managers**: Read Access (for summary reports) - **Finance Team**: Read Access (for financial and transaction-level data	Medium (depends on the dataset sensitivity)’

		review)	